

Tarasankar Das

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SUMMARY

Data analyst with an M.S. in Data Science and Ph.D. in Biophysical Chemistry, applying **SQL, Python, R, Excel, and statistical analysis** to complex datasets. Experienced in **data cleaning, reporting, visualization, dashboard-ready insights, automation-aware workflows, and stakeholder communication**.

CORE TECHNICAL SKILLS

Data Analysis & Reporting: SQL, Python, R, Excel, data cleaning, exploratory data analysis, ad hoc analysis, trend analysis, data quality review

Dashboards & Visualization: Power BI awareness, Tableau awareness, matplotlib, seaborn, reporting dashboards, KPI reporting, stakeholder-ready visualizations

Programming & Automation: Python, pandas, NumPy, Matplotlib, reproducible workflows, report automation awareness, data transformation

Statistics & Modeling: Descriptive statistics, hypothesis testing, regression, classification, model evaluation, business insights

Data Management: Data extraction, data manipulation, ETL awareness, data warehousing awareness, structured datasets, documentation, DataBricks system awareness, AWS system awareness

Communication: Cross-functional collaboration, presentation of findings, technical writing, translating complex data into actionable recommendations

SELECTED PROJECTS

Customer churn prediction using ANN | Deep Learning

- Built a 45-feature churn model (OHE + scaling) and trained a TensorFlow/Keras ANN with early stopping + softened class weights; achieved ROC-AUC 0.85, PR-AUC 0.66, and improved minority-class detection (recall 0.70). [GitHub](#)

Asthma Drug Effectiveness — ML + Statistics

- Built a feature-engineered ML pipeline on 20k patients' data with 18 features- yielding calibrated risk scores; tuned XGBoost Poisson model achieving a test MAE of 0.2746; 10% improvement over other linear regression models. [GitHub](#)

Diabetes Risk Factors — Multivariate Statistics

- Performed Welch t-tests, MANOVA, correlations with effect sizes and CIs; identified 4 key factors for prevention insights.

Macroeconomic Drivers of S&P 500 Returns (ML + Time Series)

- Built a supervised model using macro indicators (rates, unemployment, CPI, term/spread, PMIs); engineered lags/rolling features, prevented leakage, and benchmarked vs. baselines; delivered a scenario analysis.
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EXPERIENCE

Visiting Scholar | [Lehigh University, Bioengineering](#), Bethlehem, PA

March 2026 – Present

- Analyze proteomics/gene datasets using R/RStudio, Python, and Git-based workflows for pathway enrichment, STRING network analysis, and ECM biomarker prioritization.

Postdoctoral Research Associate | [University of Wisconsin- Madison](#), Madison, WI

December 2017 - January 2019

- Designed biophysical experiments and automated 2D-IR kinetic modeling for peptide structure and sequence analysis.

Postdoctoral Fellow | [Indian Institute of Technology-Bombay](#), Mumbai, India

February 2016 – November 2017

- Analyzed spectroscopy data of protein structure with feature extraction, multivariate models, & reproducible pipelines.

PhD Researcher | [Indian Institute of Science Education & Research-Kolkata](#), India

January 2010 – June 2015

- Designed multi-factor studies, built data workflows to optimize nano-probe performance in bio-microenvironments.
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SELECTED PUBLICATIONS

- S. Bhunia, P. P., and **T. Das**, Therapeutic apps of metal nanoclusters, [Luminescent Metal Nanoclusters](#), 2022, 493- 516.
- **T. Das et al.**, NIR Au nanoclusters are pH sensitive: A biomarker, [Nanomedicine](#) 12, 2016, 1105- 1112.

For complete list of publications: [Tarasankar Das \(Google Scholar\)](#)

LEADERSHIP

Graduate Student Senate Rep., Lehigh MSDS 2024–25; **V.P.**, Env. Club; **Coordinator**, Student Induction (CEC) 2019–2024

AWARDS & GRANTS

CSIR NET Scholarship 2010-2015 for **JRF, SRF**; Travel Grant award from **USA SERB; SERB-IUSSTF** Indo-US fellow 2017.

EDUCATION

[Lehigh University](#), Bethlehem, PA

July 2024 – December 2025

M.S. in Data Science; GPA: 3.76 / 4.00

[IISER-Kolkata](#), Mohanpur, India

January 2010 – June 2015

Ph.D. in Chemistry (Analytical/Biophysical)